EXERCISE-I (Conceptual Questions)

Build Up Your Understanding

GRAHAM'S LAW OF DIFFUSION

1. Which pair of the gaseous species diffuse through a small jet with the same rate of diffusion at same P and T :
(1) NO, CO (2) NO, CO_2
(3) NH_3,PH_3 (4) NO, C_2H_6
IG0001

2. The rate of diffusion of methane at a given temperature is twice that of a gas X. The molecular weight of X is :
(1) 64 (2) 32 (3) 4.0 (4) 8.0
IG0002

3. The increasing order of effusion among the gases, H_2, O_2, NH_3 and CO_2 is –
(1) H_2, CO_2, NH_3, O_2 (2) H_2, NH_3, O_2, CO_2
(3) H_2, O_2, NH_3, CO_2 (4) CO_2, O_2, NH_3, H_2
IG0003

4. Gas A having molecular weight 4 diffuses thrice as fast as the gas B at a given T. The molecular weight of gas B is :
(1) 36 (2) 12 (3) 18 (4) 24
IG0004

5. Four rubber tubes are respectively filled with H_2, O_2, N_2 and CO_2. The tube which will be deflated first is :
(1) H_2 filled tube
(2) O_2 filled tube
(3) N_2 filled tube
(4) CO_2 filled tube
IG0005

6. A balloon filled with methane CH_4 is pricked with a sharp point and quickly plunged into a tank of hydrogen at the same pressure. After sometime the balloon will have :
(1) Enlarged
(2) Collapsed
(3) Remain unchanged in size
(4) Ethylene (C_2H_4) inside it
IG0006

7. Rate of diffusion of hydrogen is :
(1) Half of He (2) 1.4 times of He
(3) Double than He (4) four times of He
IG0007

8. A football bladder contains equimolar proportions of H_2 and O_2. The composition by mass of the mixture effusing out of punctured football is in the ratio (H_2 : O_2)
(1) 1 : 4 (2) $2\sqrt{2}$: 1
(3) 1 : $2\sqrt{2}$ (4) 4 : 1
IG0008

9. If the vapour densities of methane & oxygen are in the ratio 1 : 2, the ratio of rate of diffusion of O_2 & CH_4 is respectively
(1) 1 : 2 (2) 1 : 1.414
(3) 2 : 1 (4) 1.414 : 1
IG0009

10. A gas X diffuses three times faster than another gas Y the ratio of their densities i.e., D_x : D_y is
(1) 1/3 (2) 1/9 (3) 1/6 (4)1/12
IG0010

11. The relative rate of diffusion of a gas (Mol wt. = 98) as compared to hydrogen will be :
(1) 1/7 (2) 1/5 (3) 1/4 (4) 1
IG0011

12. The relative rate of diffusion of a gas (molecular weight = 128) as compared to oxygen is
(1) 2 times (2) 1/4 times
(3) 1/8 times (4) 1/2 times
IG0012

13. Since the atomic weights of carbon, nitrogen and oxygen are 12, 14 and 16 respectively, among the following pairs of gases, the pair that will diffuse at the same rate is :
(1) Carbon dioxide and nitrous oxide
(2) Carbon dioxide and nitrogen dioxide
(3) Carbon dioxide and carbon monoxide
(4) Carbon dioxide and nitric oxide
IG0013

14. A bottle of dry ammonia and a bottle of dry hydrogen chloride connected through a long tube are opened simultaneously at both ends, the white ammonium chloride ring first formed will be :
(1) at the centre of the tube
(2) near the hydrogen chloride bottle
(3) near the ammonia bottle
(4) throughout the length of the tube
IG0014

ALLEN Session : 2020-21

15. 50 ml of a gas A diffuse through a membrane in the same time as for the diffusion of 40 ml of a gas B under identical pressure and temperature conditions. If the Molecular weight of A = 64, that of B would be :

(1) 100 (2) 250 (3) 200 (4) 80

IG0015

16. If rate of diffusion of A is 5 times that of B, what will be the density ratio of A and B :

(1) 1/25 (2) 1/5 (3) 25 (4) 5

IG0016

17. 50 ml of hydrogen diffuses through a small hole from vessel in 20 minutes time. Time taken for 40 ml of oxygen to diffuse out under similar conditions will be :

(1) 12 min. (2) 64 min (3) 8 min (4) 32 min

IG0017

18. The densities of two gases are in the ratio of 1 : 16. The ratio of their rates of diffusion is :

(1) 16 : 1 (2) 4 : 1 (3) 1 : 4 (4) 1 : 16

IG0018

19. The rate of diffusion of a gas having molecular weight just double of nitrogen gas is 56 ml per sec the rate of diffusion of nitrogen gas will be :

(1) 79.19 ml/sec. (2) 112 ml/sec
(3) 56 ml/sec (4) 90 ml/sec

IG0019

20. Under identical conditions of temperature and pressure the ratio of the rates of effision of O_2 and CO_2 gases is given by :

(1) $\frac{\text{rate of effusion of oxygen}}{\text{rate of effusion of } CO_2} = 0.87$

(2) $\frac{\text{rate of effusion of oxygen}}{\text{rate of effusion of } CO_2} = 1.17$

(3) $\frac{\text{rate of effusion of oxygen}}{\text{rate of effusion of } CO_2} = 8.7$

(4) $\frac{\text{rate of effusion of oxygen}}{\text{rate of effusion of } CO_2} = 0.117$

IG0021

DEVIATION FROM IDEAL GAS BEHAVIOUR

21. When does a real gas show behaviour same as ideal gas:

(1) At low temperature and low pressure
(2) At high temperature and high pressure
(3) At low temperature and high pressure
(4) At high temperature and low pressure

IG0022

22. In van der Waal's equation of state of the gas law, the constant 'b' is a measure of :

(1) intermolecular repulsions
(2) intermolecular attraction
(3) volume occupied by the molecules
(4) intermolecular collisions per unit volume

RG0023

23. Pressure of real gas is less than the pressure of ideal gas because :

(1) No. of collisions increases
(2) Definite shape of molecule
(3) K.E. of molecule increases
(4) Inter molecular forces

IG0025

24. Which gas can be easily liquefied ?
given 'a' for $NH_3 = 4.17$, $CO_2 = 3.59$, $SO_2 = 6.71$, $Cl_2 = 6.49$

(1) NH_3 (2) Cl_2
(3) SO_2 (4) CO_2

RG0026

25. At relatively high pressure, van der waals' equation reduces to :

(1) $PV = RT$ (2) $PV = RT + \frac{a}{V}$

(3) $PV = RT + Pb$ (4) $PV = RT - \frac{a}{V^2}$

RG0027

26. A real gas most closely approaches the behaviour of an ideal gas at :

(1) 15 atm and 200 K (2) 1 atm and 273 K
(3) 0.5 atm and 500 K (4) 15 atm and 500 K

RG0028

27. The compressibility factor of an ideal gas is :

(1) 0 (2) 1 (3) 2 (4) 4

IG0029

28. The compressibility of a gas is less than unity at STP therefore :

(1) $V_m > 22.4$ L (2) $V_m < 22.4$ L
(3) $V_m = 22.4$ L (4) $V_m = 44.8$ L

RG0030

29. The values of van der Waal's constant 'a' for the gases O_2, N_2, NH_3 and CH_4 are 1.360, 1.390, 4.170 and 2.253 L^2 atm mol^{-2} respectively. The gas which can most easily be liquefied is :

(1) O_2 (2) N_2 (3) NH_3 (4) CH_4

RG0031

ALLEN Session : 2020-21

30. Which of the following is not correct for real gases ?

(1) No force of attraction between the molecules of a gas.

(2) Volume of the moelecules of a gas is negligible in comparison to the space occupied by the gas.

(3) Do not follow PV = nRT.

(4) Both (1) & (2)

RG0054

31. The Boyle temperature or Boyle point is :

(1) The temperature at which ideal gas obeys real gas laws

(2) The temperature at which a real gas obeys ideal gas laws over an appreciable range of pressure

(3) It does not depend on nature of gas

(4) The temperature at which gases have no deviation

IG0055

ALLEN
Session : 2020-21

EXERCISE-I (Conceptual Questions) ANSWER KEY

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Ans.	**4**	**1**	**4**	**1**	**1**	**1**	**2**	**1**	**2**	**2**	**1**	**4**	**1**	**2**	**1**
Que.	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Ans.	**1**	**2**	**2**	**1**	**2**	**4**	**3**	**4**	**3**	**3**	**3**	**2**	**2**	**3**	**4**
Que.	31														
Ans.	**2**														

EXERCISE-III (Analytical Questions) — Master Your Understanding

1. When r, P and M represent rate of diffusion, pressure and molecular mass, respectively, then the ratio of the rates of diffusion (r_A/r_B) of two gases A and B, is given as :-

(1) $(P_A/P_B)^{1/2}\,(M_A/M_B)$ (2) $(P_A/P_B)\,(M_B/M_A)^{1/2}$

(3) $(P_A/P_B)^{1/2}\,(M_B/M_A)$ (4) $(P_A/P_B)\,(M_A/M_B)^{1/2}$

IG0043

2. The compressibility factor for a real gas at high pressure is :-

(1) $1-\frac{pb}{RT}$ (2) $1+\frac{RT}{pb}$

(3) 1 (4) $1+\frac{pb}{RT}$

RG0044

3. If Z is a compressibility factor, van der Waal's equation **at low** pressure can be written as :

(1) $Z = 1 - \frac{Pb}{RT}$ (2) $Z = 1 + \frac{Pb}{RT}$

(3) $Z = 1 + \frac{RT}{Pb}$ (4) $Z = 1 - \frac{a}{VRT}$

RG0045

4. Under which of the following conditions, a gas deviates most from the ideal behaviour ?

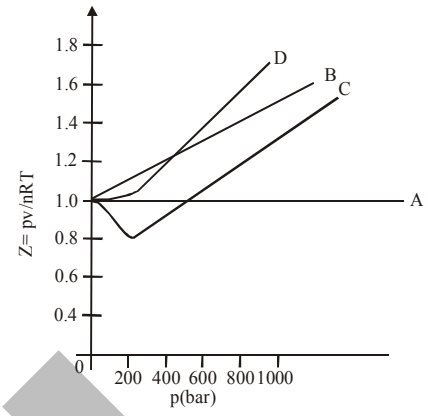
(a) Very low pressure (b) High pressure

(c) Low temperature (d) High temperature

(1) a, d (2) b, c (3) a, c (4) b, d

RG0059

5. Which curve in the graph always show compressibility factor greater than 1 ?



(1) A gas (2) B gas

(3) C & D gases (4) D gas

RG0060

6.

Gases	:	H_2	N_2	NH_3	CO_2
Critical temperature	:	33.2 K	126 K	405.5 K	304.1 K

From the above data what would be the decreasing order of liquefaction of these gases ?

(1) NH_3, CO_2, N_2, H_2 (2) CO_2, NH_3, H_2, N_2

(3) NH_3, N_2, H_2, CO_2 (4) H_2, N_2, CO_2, NH_3

RG0061

7. Vander Waals constant 'a' is -

(1) Measure of magnitude of inter molecular forces within the gas.

(2) Independent of temperature and pressure.

(3) Having unit = L^2 atm mol^{-2}

(4) All are correct

RG0062

EXERCISE-III (Analytical Questions) — ANSWER KEY

Que.	1	2	3	4	5	6	7
Ans.	**2**	**4**	**4**	**2**	**2**	**1**	**4**

ALLEN Session : 2020-21